

D10. Test, Validation, and Deployment Pack

Overview

The Test, Validation and Deployment Pack explains how to run the FlowCore project and how to confirm that each layer returns the expected result. It is written as a practical run guide for the final project package.

Local Runtime Validation

The local Bash runtime should be tested first because it creates the CSV evidence used by the C++ component.

```
cd /mnt/c/Users/<username>/Desktop/FlowCore_Master/project
bash scripts/setup_demo.sh
bash scripts/run_demo.sh
bash scripts/validate_demo.sh
bash scripts/run_cpp_priority_checker.sh
```

*The “bash” command is needed in case the user has not been given permissions using chmod. In our screenshots, it is omitted, since we have given permissions to ourselves from the early stages in our workstations.

*One should not enter the scripts directory and execute the scripts from there (e.g. cd scripts followed by ./setup_demo.sh). This is because a duplicate runtime folder will then appear inside the scripts folder, which is not correct according to the intended file structure for the project.

The expected local result is that three pallets are stored, nine status events are logged, no capacity violations are found, and the C++ priority placement rule passes.

Database Validation

The Oracle scripts should be run in this order:

```
project/sql/MySchema.sql
project/sql/Sample_Data.sql
project/sql/SQL_Operations.sql
project/plsql/PLSQL_Logic.sql
```

The SQL validation queries should not return problem rows. The PL/SQL function should compile successfully and return the expected available slots for the current warehouse locations.

Expected Validation Results

- **Bash runtime:** validate_demo.sh should report 3 pallets stored, 9 events logged, and no capacity violations.
- **SQL:** validation queries should find no stored pallets without locations, no over-capacity locations, and no orphan status events.

- **PL/SQL:** GET_LOCATION_AVAILABLE_SLOTS should return expected available slots and compile as VALID.
- **C++:** run_cpp_priority_checker.sh should check 2 priority pallets and return a passed priority placement decision.

Confirmed Local Output

```
panpontik@Yamato:~$ cd /mnt/c/Users/tsgam/Desktop/FlowCore_Master
panpontik@Yamato:/mnt/c/Users/tsgam/Desktop/FlowCore_Master$ cd project
panpontik@Yamato:/mnt/c/Users/tsgam/Desktop/FlowCore_Master/project$ scripts/setup_demo.sh
[FLOWCORE] Runtime environment prepared.
panpontik@Yamato:/mnt/c/Users/tsgam/Desktop/FlowCore_Master/project$ scripts/run_demo.sh
[FLOWCORE] Demo run completed.
panpontik@Yamato:/mnt/c/Users/tsgam/Desktop/FlowCore_Master/project$ scripts/run_cpp_priority_checker.sh
[FLOWCORE_CPP] Priority pallets checked: 2
[FLOWCORE_CPP] Priority placement decision passed.
panpontik@Yamato:/mnt/c/Users/tsgam/Desktop/FlowCore_Master/project$ scripts/validate_demo.sh
[FLOWCORE] Validation passed: 3 pallets stored, 9 events logged, no capacity violations.
panpontik@Yamato:/mnt/c/Users/tsgam/Desktop/FlowCore_Master/project$ |
```

Function GET_LOCATION_AVAILABLE_SLOTS compiled				
LOCATION_I	ZONE_CODE	CAPACITY	USED_SLOTS	AVAILABLE_SLOTS
A01	INBOUND_A	1	1	-1
A02	INBOUND_A	1	1	-1
A03	INBOUND_A	1	1	-1
B01	RESERVE_B	2	0	2
C01	HOLDING_C	1	0	1

OBJECT_NAME	OBJECT_TYPE	STATUS
GET_LOCATION_AVAILABLE_SLOTS	FUNCTION	VALID

Deployment Notes

- Run all Bash commands from the project root.
- Keep the folder structure unchanged: config, scripts, runtime, sql, plsqli, cpp, and docs.
- Run D05 before D09 so the C++ component has current runtime CSV files.
- Run the Oracle files in the documented order because each file depends on the schema and sample data being present.

Known Limits

FlowCore is a compact commissioning simulation, not a full warehouse management system. The dataset is intentionally small so the team can explain and verify every result during the presentation.